

Bridging biology and robotics: advancing submersible technology from robotic to live-fish models

Xinhao Shao^{1,2}, Jie Yang^{3,4}, Mohamad Sawan^{1,3,4}, Weicheng Cui^{1,2,4,*} 

Academic Editor: Daniel Tang Kuok Ho

Abstract

Marine resource exploration and environmental monitoring are essential for the sustainable development of oceans. However, the high cost of conventional submersibles has hindered their widespread application. In this paper, we evaluate the performance and cost of second and third generations of submersibles and propose a novel fourth-generation submersible concept based on an advanced real-time fish monitoring platform. Through a comprehensive SWOT (strengths, weaknesses, opportunities, and threats) analysis, we conduct a systematic comparison between the third and fourth generations of submersibles, highlighting their respective advantages and limitations. Additionally, the research identifies and examines the critical technical and operational challenges associated with the implementation of fourth-generation submersible platforms. This may bridge the gap from robotics to biology.

Keywords: *submersible development direction, ocean environment monitoring system, brain–computer interface, live fish*

Citation: Shao X, Yang J, Sawan M, Cui W. Bridging biology and robotics: advancing submersible technology from robotic to live-fish models. *Academia Engineering* 2025;2. <https://doi.org/10.20935/AcadEng7721>

1. Introduction

Vast ocean space and rich ocean resources can play an important role in solving human beings' three major challenging issues in the 21st century: population growth, resource shortage, and environmental pollution. The ocean has become a key scientific research field in addressing global challenges [1]. Existing submersibles are able to work in all ocean areas and at all water depths. The presently used submersibles are defined as the second-generation submersibles from a material point of view. The second-generation submersibles adopted advanced lightweight titanium alloys for pressure hull construction and implemented solid buoyancy materials for flotation capabilities, marking a significant technological advancement from the first generation, which primarily relied on conventional high-strength steel for hull fabrication and utilized gasoline systems for buoyancy provision [2]. The main problem with this generation of submersibles is its size, which is too large, resulting in its poor maneuverability. In the design of the second generation of submersibles, this problem was solved, the size became smaller, and their maneuverability was greatly improved.

However, when comparing the performances of these second-generation submersibles with real fish, the differences were still big. This is mainly reflected in agility, speed, and environmental impact. In particular, the traditional propeller propulsion mode produced a high-pitched noise, threatening marine organisms. To solve this problem and to decrease the data acquisition cost of submersibles, it was necessary to develop the third generation of

bionic fish submersibles [3]. Many research studies have been dedicated to the development of various robotic fish-type submersibles [4, 5]. However, from an economic perspective, the cost of using robotic fish is still high. In this paper, the use of a monitoring system based on real fish is proposed.

The ocean covers 71% of the Earth's surface, and the use of current second-generation submersibles and potential third-generation submersibles remains a significant economic burden. However, there are many important reasons to monitor the marine environment of whole oceans. These include at least the following five aspects: (1) Ecosystem Health: Monitoring the marine environment helps us assess the health of marine ecosystems, including the diversity of species, habitats, and the overall ecosystem [6]. This information is crucial for conservation and management efforts to protect marine biodiversity. (2) Climate Change Impact: Monitoring the marine environment allows us to track the impact of climate change on oceans, such as rising sea temperatures, ocean acidification, and sea level rise [7]. Understanding these changes helps us develop strategies to mitigate and adapt to the effects of climate change. (3) Pollution Control: Monitoring helps us identify sources of pollution in the marine environment [8], such as plastic debris, chemical contaminants, and oil spills. By monitoring pollution levels, we can take action to reduce pollution and protect marine life and habitats. (4) Fishery Management: Monitoring marine environments is essential for sustainable fishery management [9].

¹Research Center for Industries of the Future, Westlake University, Hangzhou 310030, China.

²Key Laboratory of Coastal Environment and Resources of Zhejiang Province, School of Engineering, Westlake University, Hangzhou 310030, China.

³CenBRAIN Neurotech, Westlake University, Hangzhou 310030, China.

⁴Integrated-on-Chips Brain-Computer Interfaces Zhejiang Engineering Research Center, Hangzhou 310030, China.

*email: cuiweicheng@westlake.edu.cn

It helps assess fish stocks, track fishing activities, and enforce regulations to prevent overfishing and protect marine resources for future generations. (5) Human Health: Monitoring the marine environment is important for public health as it helps identify potential risks from contaminated seafood, harmful algal blooms, and waterborne diseases. Monitoring ensures the safety of seafood consumption and recreational activities in marine waters. Overall, monitoring the marine environment is essential for understanding, protecting, and sustaining the health and productivity of our oceans, as well as ensuring the well-being of marine ecosystems and human populations that depend on them [10]. How to dramatically reduce the construction and operational costs of submersibles is always the key issue for realizing the environmental monitoring of whole oceans.

The purpose of this paper is to provide revolutionary thinking for the reduction in submersible construction and operational costs for ocean environment monitoring systems, and this is defined as the fourth-generation submersibles. A SWOT analysis was carried out to compare the third and fourth generations of submersibles, considering their strengths, weaknesses, opportunities, and threats.

2. Submersible application cost analysis

Cost is the main issue for monitoring systems for marine environments, especially for various submersibles, which are the core equipment for ocean exploration. Traditional cost reduction methods include the following: (1) Standardization: Designing and manufacturing standardized components for submersibles can help lower production costs by streamlining manufacturing processes and reducing customization expenses. (2) Redesign: Simplifying the design of submersibles by focusing on essential functionalities and minimizing complex features can reduce manufacturing costs and maintenance requirements. (3) Material Selection: Choosing cost-effective yet durable materials for submersible construction lowers production costs without compromising performance and durability. (4) Modular Design: Implementing a modular design approach allows easier assembly, maintenance, and upgrades of submersibles, reducing overall costs associated with repairs and replacements. (5) Use of Commercial Off-the-Shelf (COTS) Components: Incorporating readily available COTS components in submersible systems can help reduce development costs and shorten maintenance times compared to custom-made components. (6) Collaboration and Partnerships: Collaborating with research institutions, industry partners, and government agencies can help share resources, expertise, and costs associated with submersible development and deployment. (7) Automation and Remote Operation: Implementing automation features and remote operation capabilities in submersibles can reduce the need for human intervention, lowering operational costs and improving efficiency. (8) Energy Efficiency: Designing submersibles with energy-efficient systems, such as low-power sensors and propulsion systems, can reduce operational costs associated with power consumption and battery usage.

These traditional cost-saving strategies have already been implemented in the currently used second-generation submersibles and are also suitable for the third-generation robotic fish-type submersibles. However, from the perspective of a new definition of task efficiency [11], one can find that if we can install the sensors directly into a living fish and use the brain-computer interface

(BCI) to control the fish according to human will, both the construction and operational costs could be significantly reduced. In this section, we will briefly introduce the concept of task efficiency, which is suitable for comparing the real costs of both submersibles and live fish.

At present, there is no very appropriate equation to describe the swimming efficiency of fish; many researchers have compared it with the propulsive efficiency of a submersible, according to Equation (1). The accuracy of this calculation has been questioned, and this depends on the oversimplification of metabolic rate, thrust, resistance, etc.

$$E_{\text{swim}} = E_{\text{propulsion}} = (W_{\text{useful}}/C) \times 100\% \quad (1)$$

Here, E_{swim} means swimming efficiency, $E_{\text{propulsion}}$ means propulsion efficiency, W_{useful} means useful propulsion work, and C means input energy consumed. Useful propulsive work here refers to a kind of interactive energy, specifically the energy transferred by a submersible or a fish to its motion carrier in the water. The calculation of this energy mainly depends on the comprehensive calculation of its motion trajectory, velocity, and resistance measurement. While this approach makes sense for submersibles, it is less applicable for real fish because they need to think about feeding, which creates a lot of uncertainty [12–16].

As a real organism, the mechanical efficiency of fish swimming is mainly determined by its body shape, bone density, and movement pattern. Even if two identical fish were to swim at the same route and time, it would be unlikely that they would consume exactly the same amount of food. The free choice of different movement patterns of fish, including static, accelerating, jumping, gliding, and other ways, leads to a great variance in their swimming efficiency calculations [17, 18].

In order to solve the challenge of measuring fish swimming efficiency, Cui et al. proposed to use the concept of task efficiency [11]. From a theoretical perspective, the propulsion and resistance of live fish exhibit inseparability, with their swimming efficiency being significantly influenced by individual autonomous behavioral regulation. When the neural system is separated from physiological functions, the organism loses its vital signs and becomes incapable of generating any effective propulsive force. This mechanism fully explains the fundamental differences in movement characteristics between live and dead fish. To overcome the limitations of traditional models in evaluating the swimming efficiency of live fish, this study innovatively proposes an assessment method that substitutes swimming efficiency with task efficiency. Cui et al. conducted a retrospective study of task efficiency for real fish and found that the field has gradually shifted from early thrust- and drag-based measurements to more accurate measurements that include energy output, useful energy, and transportation costs [11]. In terms of the cost of transport suggested by Maertens et al., they used the same unit price for P_{in} and P_{useful} , where P_{in} is the real power we paid for robotic fish, but P_{useful} is not the real power we received [15]. Cui et al. argued that these unit prices should be different, and they provided the following definition for improvement:

$$\eta_{QP} = \frac{C_{\text{useful}} P_{\text{useful}}}{C_{\text{in}} P_{\text{in}}} = \frac{C_{\text{useful}} (T_n + R_d) U_s}{C_{\text{in}} P_{\text{in}}} \quad (2)$$

where C_{useful} and C_{in} are the values estimated by us according to the current market price for the unit's useful work and the unit's input energy.

Energy efficiency ratio is a key parameter that involves a subjective evaluation. When we design vehicles in a mission-specific context, the focus is not only on controlling costs and maintaining maneuverability, but also on accomplishing the actual mission. The value of these tasks can be called the added value, so Cui et al. proposed to replace the traditional calculation method of swimming efficiency with task efficiency, which is defined according to Equation (3).

$$\eta_t = \frac{VAL_{task}}{COST_{task}} = \frac{VAL_{useful} + VAL_{added}}{COST_{direct} + COST_{shared}} \quad (3)$$

Here, VAL_{task} is the value created by performing this task using the vehicle, and it can be decomposed into two parts. One is the equivalent value created by just moving the vehicle from the initial location to the target location, VAL_{useful} is the same as Equation (2), and the other is the added value VAL_{added} created by performing tasks such as transporting other goods, measuring environmental data, sampling along the trajectory, and making a new discovery when executing this task. $COST_{direct}$ is the direct cost consumed by this task, such as electricity or fuel for robotic fish and submersibles, or food costs for a living creature, including fish. $COST_{shared}$ is the shared cost charged to this task by considering the construction and operation of the vehicle. The use of a mothership has been found to be the dominant factor for the shared cost [12].

Equation (3) is a much broader definition than Equation (2). Taking bionic fish as an example, if the added value and shared cost are removed from Equation (3), the evaluation criteria will degenerate into Equation (2). Then, if the unit cost difference between the active effort and the input energy is removed on the basis of Equation (2), Equation (2) will be equivalent to the definition of Maertens et al. [15]. Therefore, the task efficiency defined by Equation (3) is more closely related to the real concern of designers. In the design of the vehicle, our goal should be to use Equation (3) as the objective function and continue to optimize it. The nutritional input for fish can range from high-cost to low-cost feed options, with the selection significantly impacting their operational efficiency. When utilizing inexpensive feed, the efficiency of live fish may substantially outperform robotic fish systems. Equation (3) provides a comprehensive framework that accounts for all critical factors affecting task efficiency, which are applicable to both autonomous underwater vehicles and biological fish specimens. Notably, this model incorporates the substantial impact of informational inputs. Furthermore, the model acknowledges the capacity of fish trainers to modulate fish behavior through targeted instructional techniques during operational processes. Comparing live fish and bionic fish, it is found that the variables in Equation (3) can be measured experimentally. For example, motion trajectory, speed, acceleration, etc., can be measured directly by the sensor, while resistance can be approximated by the dead fish model. Although there is a difference between live and dead fish, if both are traveling in the same direction with the same speed and acceleration, they are actually carrying out the same amount of useful work. So, the key difference is the cost of food for fish versus the cost of energy for bionic fish. If the actual shape of the real fish at any time can be

recorded, by using computational fluid dynamics (CFD), the actual drag at any time can be accurately calculated. Then, by measuring the trajectory of the fish, the active force can also be deduced by using Newton's second law.

From this equation, it is obvious that if we can compare the two monitoring systems of a robotic fish type and a living fish type, the latter is much cheaper than the former. For the robotic fish, one needs the cost of a submersible in addition to the measuring system, as well as the support of the mothership for charging the battery, but for the living fish type, there is no construction cost of the submersible, and there is no need for the support of the mothership to supply the battery. The fish can survive themselves since the ocean can provide plenty of food for them. So, compared with the optimization of the performances of robotic fish, this is a revolutionary idea to develop an ocean environmental monitoring system based on real fish.

3. General concept of marine environmental monitoring systems based on live fish

The general marine environmental monitoring system, based on real fish, as shown in **Figure 1**, consists of five subsystems: (1) the marine environmental monitoring sensor system; (2) the data-recording system; (3) the information transmission system; (4) the fish control system; and (5) the autonomous decision-making system. The detailed technical issues to be addressed in realizing these five subsystems will be discussed in Section 5. In this section, we will provide an overview of the current state of technology for using marine organisms as environmental monitoring platforms, as well as the development status of animal robots.

This study aims to further improve the marine environment monitoring platforms mentioned above, which also use live marine organisms such as fish, but here, this study adds to the research on fish brain science, controlling fish movement through brain-computer interface chips to achieve the purpose of accurate monitoring. So, it can be seen as a combination and enhancement of animal robots and biotelemetry platforms. The following are related studies of marine animal robots: Among marine vertebrates, fish and sea turtles are frequently utilized for environmental surveillance. Both are vertebrates, with complex motor nervous systems that are organized in a hierarchical, multi-level manner. This hierarchy comprises the cerebral cortex, basal ganglia, mesencephalic motor areas, spinal motor circuits, and skeletal muscles. Sensory information, such as the spatial position of movement targets and their relationship to body position, is processed by the cerebral cortex, thalamus, and mesencephalon. These data are then forwarded to the basal ganglia, which integrate inputs from higher levels and formulate behavioral decisions. Intermediate neurons in structures such as the globus pallidus and reticular substantia nigra modulate the activity of mesencephalic motor areas. The resulting motor commands are transmitted through the reticulospinal system to the central pattern generators (CPGs) in the spinal cord, activating networks within the spinal cord. This network, together with the reticulospinal tracts, CPGs, motor neurons, and the musculoskeletal system, executes movement.

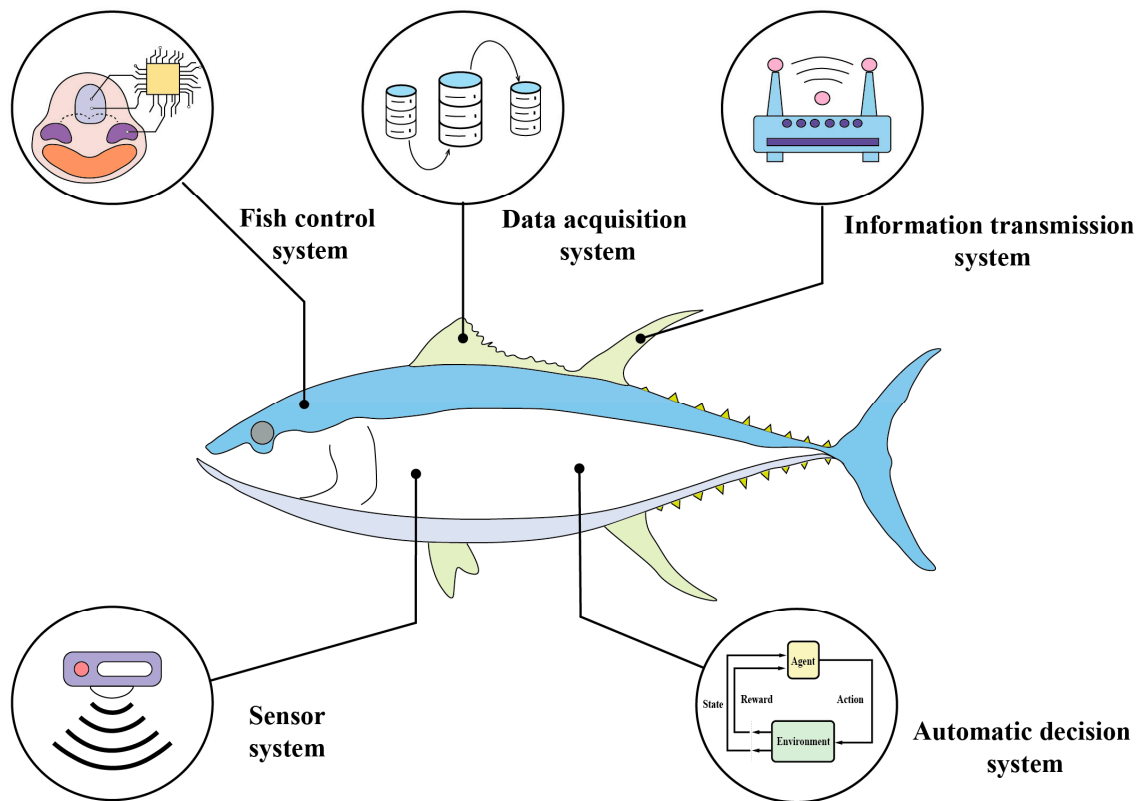


Figure 1 • Marine environment monitoring system based on live fish.

The concept of using animals as platforms to monitor the environment has been explored since 1965 [19]. Kooyman investigated the diving habits of the Weddell seal (*Leptonychotes weddelli*) by fitting the depth sensor to the animal, achieving the first instance of the marine biotech-based monitoring of environmental parameters. Subsequent researchers extended the use of marine organisms as monitoring platforms to penguins [20], sharks [21], turtles [22], and so on. **Figure 2** provides examples of biorecords and biotelemetry tags on the model organism.

Shark Robots: In their review of aquatic animal robots, Peng et al. reported that in 2015, Su et al. employed a combination of mechanical and electrical stimuli to control shark behaviors. The lateral line canal between the dorsal fins responds to mechanical stimuli, while the Lorenzini ampullae in the head are sensitive to electrical stimuli [24]. In 2006, Lehmkuhle separately implanted chips and microelectrodes in the shark's olfactory regions, using electrical stimulation to produce virtual odor stimuli that induce odor-tracking swimming behaviors [25, 26].

Sea Turtle Robots: Kim et al. designed a system that controls turtle behavior based on their ability to discern between specific wavelengths of visible light and innate avoidance behaviors. The system consists of a light stimulation system mounted on the turtle's back and a BCI-HMD device worn by the researcher. The imaging device within the light stimulation system captures images, based on which the researcher issues commands, guiding the turtle's movement and receiving image feedback for further commands [27, 28].

Carp Robots: In 2019, Peng et al. implanted stimulation electrodes in the carp's brain motor areas to control forward movement and turns [29]. In 2021, leveraging the carp's responses to different light wavelengths, they controlled the fish's direction and speed underwater using a light control device attached above the olfactory stalk, thereby avoiding complications like intracranial bleeding associated with direct brain implants [30].

Goldfish Robots: In 2009, Nobutaka et al. implanted stimulation electrodes near the nucleus of the medial longitudinal fasciculus (Nflm) in the goldfish's midbrain motor area, achieving unilateral stimulation to induce directional turning. The medial line stimulation in the Nflm induced forward movement [31]. In 2017, Vinepinsky's team developed a wireless signal acquisition system to investigate the relationship between light exposure and visual

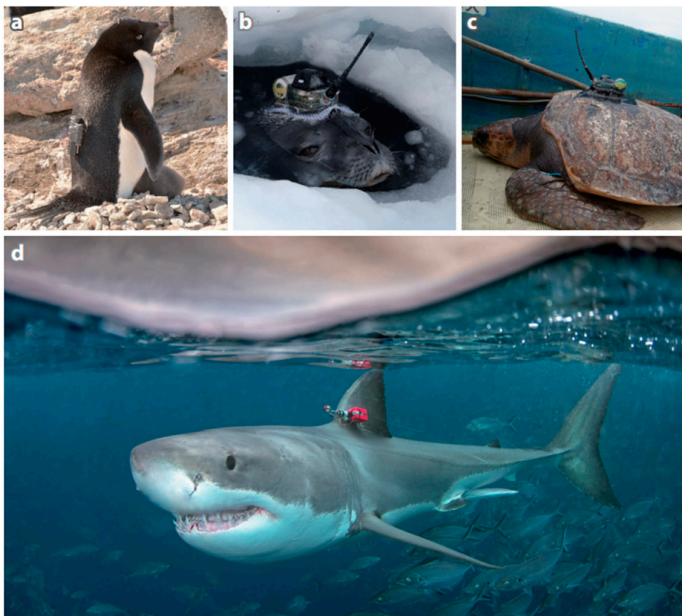


Figure 2 • Biologging and biotelemetry tags attached to (a) an Adélie penguin, (b) a Weddell seal, (c) a loggerhead turtle, and (d) a white shark. Photos courtesy of (a) Yuuki Watanabe, (b) Nobuo Kokubun, (c) Tomoko Narazaki, and (d) Andrew Fox [23].

nerve signals. They later refined this system to remotely adjust the position of the implanted electrode, allowing for the study of additional neurons involved in navigation and positioning within the goldfish brain [32, 33]. **Figure 3** is a physical picture of the goldfish robot.

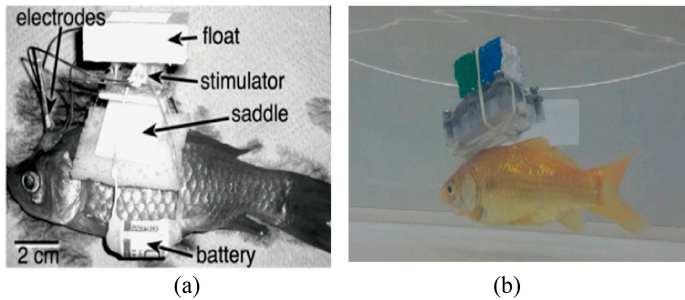


Figure 3 • Research progress of marine biological robots: (a) Kobayashi et al.'s work with goldfish robots in 2009 [31] and (b) Vinepinsky et al.'s work with goldfish robots in 2017 [33].

The development of vertebrate monitoring platforms has extended to various other animals. In 2002, Talwar developed the first rat robot, which achieved forward movement and turning by stimulating the S1 whisker representation area and the MFB (medial forebrain bundle). Stimulation of the MFB induced pleasure, while stimulation of the S1 area guided turning [34]. Between 2005 and 2007, Su et al. developed similar robots, and in 2009, they achieved movement by stimulating the VPL (ventral posterolateral nucleus) and AMY (amygdala nucleus), with VPL stimulation inducing pain to guide turning and AMY stimulation inducing fear to promote forward movement [35]. In 2016, Xu et al. at Zhejiang University achieved precise turning by stimulating the VPM (ventral posteromedial nucleus), optimizing tactile perception [36]. In 2010, Zhen et al. controlled stopping through Dipag (dorsolateral periaqueductal gray) stimulation and combined optogenetic techniques to enhance precision [37, 38]. Additionally, in 2017, Capelli et al. found that LPGi (lateral paragigantocellular nucleus) stimulation could accelerate movement, and in 2018, Caggiano discovered that CnF (cuneiform nucleus) controlled high-speed movement, while PPN (pedunculopontine nucleus) controlled low-speed movement [39, 40]. In 2008, Su et al. controlled pigeon flight by stimulating DIVA (nucleus dorsalis intermedius ventralis anterior) and the old striatum, optimizing electrode implantation and takeoff control [41]. In 2021, Wang et al. induced avoidance behavior by stimulating the PoA (preoptic area) and DIVA. In 2012, Dai et al. found that locus coeruleus stimulation triggered wing-flapping responses, and in 2015, further research revealed that FRM (formatio reticularis medialis) was associated with turning control [42, 43].

In addition to vertebrates, invertebrates, represented by a variety of insects, are also being studied as monitoring platforms. Invertebrates have simple neuromuscular systems, including the brain, thoracic and abdominal ganglia, and motor neurons. Sensory organs located in the head sense external stimuli and transmit information to the brain or thoracic ganglia via sensory neurons. The motor center then integrates these signals to generate control commands that are transmitted to motor neurons via the thoracic and abdominal ganglia to coordinate muscle movements and subsequent behavioral actions. In 2009, Bozkurt et al. proposed the early metamorphosis implantation technique (EMIT), in which electrodes were implanted in the moth pupal stage to improve stability and avoid adult tissue damage, and directional control was

achieved by stimulating antennal lobes and neck muscles [44]. In 2012, he used wireless backpack stimulation to control Madagascar hissing cockroaches to move along a specified path. In 2013, Backyard et al. launched a Cyborg Roboroach, which is controlled by a Bluetooth-enabled smartphone [45, 46]. In 2009, Sato et al. developed a wirelessly controlled backpack to achieve precise flight control by stimulating the visual lobes and flight muscles of beetles [47]. In 2017, Gu et al. designed a non-invasive system that uses LED light stimulation combined with virtual reality and camera feedback to control the flight path of bumblebees [48].

Through the in-depth analysis and summary of the existing animal robot cases, bio-inspired robot technology has shown significant advantages in the field of environmental monitoring. These cases not only demonstrate the adaptability and efficiency of biological robots in complex environments but also provide a strong theoretical support and practical reference for the construction of marine monitoring platforms. Based on this, the marine monitoring platform based on marine biological robots proposed in this paper not only inherits the flexibility, energy-saving, and strong adaptability characteristics of biological robots, but also further improves its monitoring capability and reliability in the marine environment through an optimized design.

4. SWOT comparative analysis of the third- and fourth-generation submersibles

From the definition of the fourth generation of submersibles, it can be seen that the technological progress of the first three generations is progressive, while the fourth generation is subversive. It would be helpful to compare the third and fourth generations of submersibles in an objective way. Therefore, a strength, weakness, opportunity, and threat (SWOT) analysis was carried out.

SWOT analysis has become a key tool for strategic planning [49]. Since the 1960s, SWOT analysis has been adopted by many strategic planning researchers and scholars. This type of analysis produces a more differentiated and accurate strategic decision tool that is in line with society's expectations for a safer and sustainable world [50].

The strength, weakness, opportunity, and threat analysis of the third and fourth generations of submersibles is shown in **Table 1**.

The technical characteristics of the third-generation biomimetic robot fish and the fourth-generation live fish-controlled submersibles are significantly different. In terms of strengths, the core competitiveness of the third-generation submersible is reflected in the level of technical controllability. Its fully human-designed motion logic and sensor layout support accurate programming, and its modular material optimization strategy enables it to withstand a 60 MPa deep-sea pressure environment. In contrast, the core advantage of the fourth-generation system comes from biological efficiency; the real fish carrier not only achieves 100% natural propulsion efficiency, but its metabolic system can supply energy for a long time, and the sensor's carrying cost is a fraction of that of the bionic machine, which is subversive for the construction of large-scale marine monitoring networks. The weaknesses of the two reflect the essential contrasts of the technical route. The third-generation system is limited by the physical bottleneck of bionic drive technology; its artificial muscle propulsion efficiency is only part of the biological efficiency of real fish, while the deep-sea

pressure shells are expensive, leading to commercial application. Although the fourth-generation system has cost advantages, it faces the challenges of insufficient decoding accuracy of neural signals and the potential legal liabilities related to animal welfare considerations. These risks make its large-scale deployment have institutional obstacles. From the perspective of development, the window of opportunity for the fourth-generation technology lies in bioelectronic fusion innovation combined with carbon-based degradable sensor technology, which is expected to build a zero-ecological-interference monitoring system. At the threat level, both generations of technology face serious challenges. The third-generation system’s lithium-ion battery leakage risk can cause

heavy-metal pollution, while the fourth-generation technology, if it fails to solve the sensor biocompatibility problem, may cause the abnormal mortality of fish populations. The technological evolution path shows that the third-generation system needs to focus on the breakthrough of single performances to serve in high-end scenarios such as military and industrial applications, while the fourth-generation technology should focus on the construction of distributed monitoring networks based on fish intelligence. The key to its success is to balance bioethics and technical utility. At the same time, this study will be further compared in three aspects: monitoring application, technical capability, and technical limitation.

Table 1 • The SWOT analysis results for the third and fourth generations of submersibles.

Item	Third-generation submersibles	Fourth-generation submersibles
Strengths	1. Technical controllability: There is a completely manual design and motion logic, and the sensor layout can be precisely programmed for control.	1. Biological performance advantage: The propulsion efficiency of real fish is 100%, and the energy metabolism system is self-sustaining. They have natural pressure/corrosion resistance and are adaptable to depths far beyond mechanical systems.
	2. Environmental adaptability: The material can be optimized to adapt to extreme marine environments such as high pressures and low temperatures.	2. Very low cost: Carrier costs approach zero, and maintenance costs are reduced by 80%.
	3. Ecological compatibility: The bionic shape and bionic propulsion reduce disturbance to marine life.	3. Ecological integration: Almost-natural biological behavior can obtain interference-free environmental data, and the fish network can achieve three-dimensional, large-scale coverage.
	4. Stability: The energy supply (battery) is controllable, the task cycle is predictable, and the data return link is more stable than the biological carrier.	
Weaknesses	1. High-cost bottleneck: Precision bionic drive systems, deep-sea pressure housing, and anti-corrosion materials lead to a single machine cost that is far more than the biological carrier.	1. Control of technology bottleneck: The lack of decoding accuracy of neural signals leads to the deviation of the movement path, the risk of autonomous behavior of the biological carrier, and instruction conflicts.
	2. Energy efficiency constraints: Propulsion efficiency is only 30-60% of that of real fish, and battery energy density limits battery life.	2. Ethical legal risk: Animal rights groups oppose the use of biological manipulation techniques, which may violate the relevant provisions of the animal welfare regulations.
	3. Mechanical reliability risk: The failure rate of complex mechanical structures increases in a salt erosion environment.	3. System vulnerability: The sensor is vulnerable to fish friction, parasite damage, and carrier natural death, which lead to data link interruption.
Opportunities	1. Biomimetic technology progress: Breakthroughs in flexible robots and artificial muscle technology can improve movement efficiency, and AI autonomous navigation algorithms can optimize path planning capabilities.	1. Biotechnology convergence: Symbiotic sensors (such as electronic tattoos on fish skin) reduce damage.
	2. Policy-driven developments: The localization strategy of marine equipment promotes the development of the bionic robot industry chain.	2. Marine big data needs: Low-cost intensive data are urgently needed for monitoring ocean carbon sinks in the context of carbon neutrality.
Threats	1. Substitution technology impact: Live fish control technology could disrupt the market if it breaks through ethical barriers.	1. Ethical resistance The public rejection of “animal mechanization” may hinder commercialization.
	2. Potential contamination: Lithium battery leakage and metal shell shedding of heavy-metal elements caused ecological concerns.	2. Ecological risk: Fish carrying equipment may cause pollution of the ecological chain (e.g., heavy-metal precipitation from sensors)

Monitoring applications: The third- and second-generation submersibles cannot conduct data collection in certain environments, such as deep-sea basin temperature monitoring and polar-sea environment monitoring. However, these harsh environments are not a problem for the organisms growing in them, so the academic community generally adopts biological robots for environmental monitoring in harsh environments. For example, McMahon et al. used seven *Dermochelys coriacea* turtles equipped with the most advanced satellite relay data recorder (SRDL) as the geothermal monitoring platform in the sea basin. Temperature monitoring was carried out [22]. Treasure M. et al. used a southern elephant seal with CTD-SRDL as a polar-ocean monitoring platform [51].

Technical capabilities: In terms of autonomous control, biological systems show unique advantages: biological muscles achieve micron-level continuous deformation adjustment, and their fine-scale motion control is two orders of magnitude higher than that of discrete step responses of artificial actuators. The synaptic plasticity of neural networks and the synergistic effect of glial cells give biological vectors the ability to repair themselves within 72 h, while electronic systems require manual intervention and are more expensive to maintain with the same failure rate. At the metabolic level, the multi-level conversion efficiency of the biological energy pyramid reaches 65–80%, far exceeding the limit value of the lithium drive system. These differences lead to the insufficient adaptive ability of artificial bionic bodies in complex fluid environments, and it is difficult to reduce energy consumption costs.

Technical limitations: For the third- and fourth-generation submersibles, the impact of monitoring the deep-sea environment on the sensors and communication modules cannot be ignored. The technical bottleneck of the system is mainly reflected at the sensor module level, where the deep-sea water pressure (>60 MPa) causes the nonlinear drift of the traditional MEMS pressure sensor, and the low temperature (2–4 °C) causes the abnormal fluctuation of the sensitivity coefficient of the strain gauge; in terms of communication, the underwater wireless optical communication is affected by the scattering of suspended matter, while the multi-path effect of acoustic communication leads to the random delay of control instructions.

5. Key technologies to be solved in the development of fourth-generation submersibles

From **Figure 1** of a representative marine environmental monitoring system based on real fish, at least five subsystems are needed. While the information transmission system and data-recording system are similar to those used in the second or third generations of submersibles, the other three subsystems are entirely new and developed specifically for fourth-generation submersibles. Therefore, in this section, we identify the key techniques involved in designing and developing these three subsystems.

5.1. Fish control system

A fish control system is the core component of the fourth-generation submersible technology; its main function is to achieve an accurate control of fish behavior through brain–computer interface (BCI) technology and external stimulation means. This system leverages the instinctive behavior of fish, enabling them to complete complex tasks in a specific environment while significantly

reducing the cost of building and operating the system, providing a revolutionary solution for large-scale marine environmental monitoring.

At present, the research on such aquatic animals, such as live fish, mainly focuses on brain region recognition, electrical stimulator design, experimental training, etc. Peng et al. noted that the medial longitudinal fasciculus nucleus (NFLM) in carp's midbrain affects tail movement, but this finding has not been used for robotic navigation control in carp. To bridge this gap, NFLM electrical stimulation was used to quantitatively regulate the forward and steering behavior of carp robots. Without craniotomy, brain electrodes are implanted in the NFLM, allowing controlled underwater experiments with different stimulation parameters. ImageJ software and custom programming were used to measure the forward speed and turning angle of the robot before and after the stimulus. The results show that the steering motion can be adjusted between 30 and 150 degrees, while the forward speed can reach 100 cm/s, demonstrating the precise control of the robot's movement [52].

According to this kind of research, the technical route of the brain–computer interface control of fish can be summarized in terms of brain region recognition, the design of the brain–computer interface chip, EEG signal acquisition, EEG signal processing, and stimulus signal generation.

Brain region recognition: Brain region recognition is the first step of brain–computer interface research. The purpose is to find the key brain region related to the specific motor behavior of fish [53]. At present, the research on the brain region of fish mainly focuses on the following: (1) The integration of eye and motor movements: Saitoh et al. analyzed how the brain region of fish coordinated eye movement and body movement [54]. (2) The regulatory role of reticulospinal neurons: Zelenin et al. found that reticulospinal neurons in lamellar eels control forward and backward swimming by regulating the frequency and intensity of neuron firing [55]. (3) The navigation and behavioral guidance of the parietal lobe: Saitoh et al. found that fish can integrate eye, body orientation, and motor movements by stimulating the tectum, which induces motor control, for goal-directed behavior [54]. (4) The influence of the cerebellum on the motor rhythm: Green et al. found that in zebrafish, the cerebellum affects swimming speed and direction by regulating neuronal activity [56]. (5) The feedback regulation of the lateral line system: Roberts et al. found that lateral neurons in fish have a protective effect on motor stimulation, preventing the sensory organs from being overstimulated [57].

Design of brain–computer interface chip: Brain–computer interface (BCI) chip design is an interdisciplinary research field combining neuroscience, microelectronics, and computing science, with the goal of establishing a direct communication channel between the brain and external devices. Modern BCI chip systems usually include front-end neural signal acquisition, back-end electrical stimulation, signal processing, wireless communication, and energy management modules, and gradually develop towards high integration, miniaturization, and low power consumption. For example, flexible microelectrode arrays and high-channel-count systems, such as the 3072 electrode thread system developed by Neuralink, significantly improve signal acquisition accuracy and long-term implantation feasibility [58]. At the same time, closed-loop control algorithms, such as ReFIT Kalman filters, optimize real-time signal processing capabilities and improve the sensitivity and stability of device responses [59]. In addition, low-power

design technologies and wireless power supply technologies (such as WiNoC architecture and implantable antennas) provide key support for wearable devices and long-term implantation [60]. At the same time, there are many brain–computer interface chip designs suitable for implantation in fish. An example of this was created by Mercanzini et al.; their micromachining device is made of polyimide and platinum and offers mechanical flexibility, high-quality electrical characteristics, and excellent biocompatibility. The chip integrates a 32-channel platinum–iridium alloy micro-electrode array with a diameter of 5 μm and a spacing of 50 μm , supporting the simultaneous collection of local field potential (LFP) and single-neuron action potentials. In order to reduce tissue damage, the electrode surface is coated with a PEDOT:PSS conductive polymer coating. Its impedance is stable in the range of 50–100 k Ω (1 kHz). Inside the chip is a low-power Bluetooth module (power < 1.2 mW) [61].

EEG signal acquisition: EEG acquisition is a key step in the brain–computer interface (BCI) system, which establishes the communication channel between the brain and external devices by recording the electrical signals generated by the activity of brain neurons. Signal acquisition can be divided into invasive and non-invasive methods: invasive methods include cortical electrodes (ECoG) and deep brain recording, which can obtain high-resolution signals, but require surgical implantation, and are suitable for high-precision neural manipulation scenarios; non-invasive methods such as electroencephalography (EEG), magnetoencephalography (MEG), and near-infrared spectroscopy (NIRS) are widely used in research and other applications, especially for broad populations and early exploration due to their safety and convenience [62]. Modern EEG acquisition devices incorporate highly sensitive electrodes and front-end amplifiers that are capable of recording weak neural signals without introducing significant noise. For example, flexible electrode arrays improve comfort and stability when recording over long periods of time [63]. At the same time, low-power-consumption and wireless transmission technologies are widely used, allowing devices to be lightweight and operated remotely [64].

EEG signal processing: The collected EEG signals need to be pre-processed to reduce artifacts and noise, including filtering, de-baseline drift, and artifact elimination, in order to extract useful neural activity features. Combined with machine learning algorithms, EEG acquisition systems can translate neural activity patterns in real time, supporting a variety of applications such as neuroprosthesis control, communication assistance, and brain function research.

Stimulus signal generation: Electrical stimulation signal generation plays an important role in the brain–computer interface (BCI) system to drive motor function, especially in the study of the motor control of fish. By precisely adjusting the intensity, frequency, and timing of electrical stimulation signals, specific motor control areas of the nervous system of fish can be activated, thereby guiding or inducing the motor response of fish. This process involves the precise regulation of motor neural networks, especially through the electrical stimulation of specific areas of the spinal cord or brain, to achieve the regulation of motor behavior. In fish movement studies, electrical stimulation is usually carried out by implanting microelectrode arrays or external electrodes, and the choice of stimulation parameters (such as frequency, waveform, duration, etc.) directly affects the effect of stimulation and the fish's motor response. Zhao et al. found that the abductive nucleus (NVI) is

a motor nucleus that controls ipsilateral steering, and the facial nucleus (NVII) is a motor nucleus that controls forward motion. By adjusting the stimulation voltage and the number of stimulation pulses, the carp can be controlled stably, and the left–right steering movement of 30–180° and the forward movement of more than 80 cm/s can be achieved. The electrical stimulation signal used in this study is shown in **Figure 4**. The positive and negative pulse square waves of different frequencies are shown in **Figure 4**; the amplitude is ± 5 V, the electrical stimulation pulse is 5 ms, and the stimulation frequency is changed to 1 Hz, where the lower limit is 1 Hz and the upper limit is 60 Hz [65].

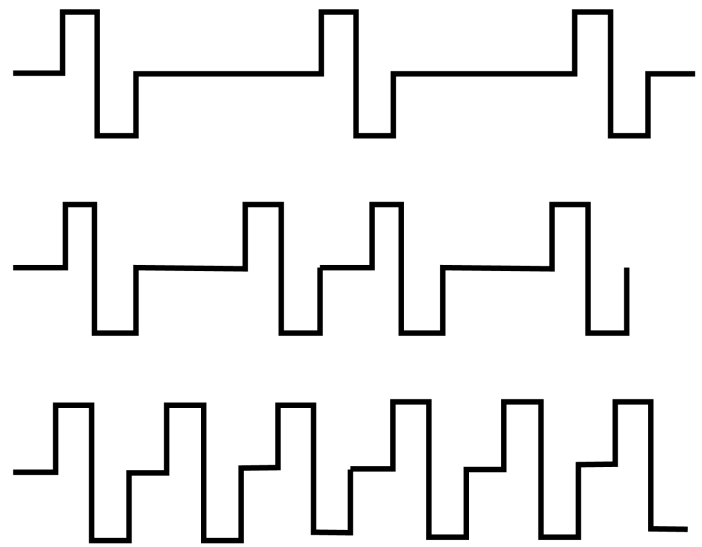


Figure 4 • Square waves of positive and negative stimuli of different frequencies.

It is worth mentioning that, in general, the waveform of the electrical stimulation signal is not the cause of a direct influence on the motion parameters. Especially in brain–computer interface control experiments, the choice of motor nuclei in the brain area is the decisive factor in determining whether the stimulation can be successfully induced.

Bioimplant surgery: In this study, we summarize the general procedural approach to brain electrode implantation in an aquatic animal model based on studies in the literature, using carp surgery as an example. Surgery usually involves the following steps: First, the animal is anesthetized to ensure it remains stable during the procedure (here, a 0.36 g/L solution of eugenol is usually used). Next, the animal is carefully positioned with a brain locator, and the appropriate implant channel is selected to access the target brain region. The electrodes are then implanted using precise stereotactic techniques to ensure accurate placement. Postoperative care, including monitoring and potential anti-inflammatory therapy, is recommended to support recovery and reduce complications.

For example, in the carp implantation protocol by Zhao et al., the fish was anesthetized with eugenol solution and securely anchored in a stereotactic device. Brain electrode implantation channels were selected in the medulla oblongata region, and the distance between adjacent channels was greater than 1 mm. The self-made electrodes were implanted into the medulla oblongata using a stereotactic device. This method can precisely control the underwater motor behavior of carp by adjusting the electrical stimulation parameters targeting specific motor nuclei (NVI and NVII) in the medulla oblongata [65].

When using brain–computer interface technology to control the movement of fish, the main challenges are as follows:

Brain region recognition: The research accuracy of the brain region of fish is still low, and it is impossible to accurately map the specific amplitude of a specific movement to the brain region.

Electrical stimulation and stress response: The brains of fish are fragile, and too much current will cause a stress response in fish, which is not conducive to their control. And too small a current cannot cause a large enough amplitude of action.

Equipment miniaturization: For the current research on animal

robots, most of them still use larger EEG equipment, and for fish, the use of larger EEG equipment not only requires strict water-proofing of the equipment, but also greatly affects the movement of fish.

Fish training costs: As the carrier of the fourth-generation submersible, live fish can indeed greatly save on operation and maintenance costs, but at the same time, it is necessary to consider the training cost of fish, that is, training a fish with a brain–computer interface into a fish that can complete the task. This is mainly based on the precise identification of brain regions and the durability of the brain–computer interface. The technology roadmap is shown in **Figure 5**.

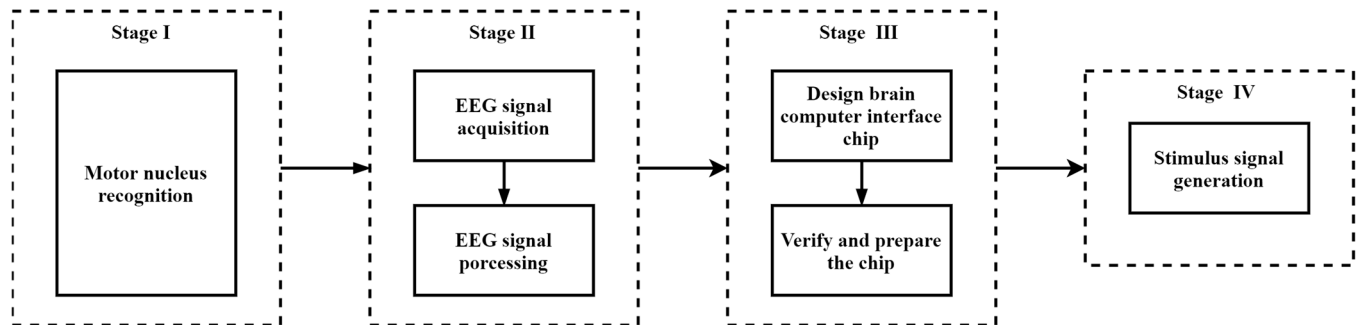


Figure 5 • Technical roadmap for brain–computer interface control of fish.

5.2. Autonomous decision system

In the process of using real fish for monitoring tasks, the operator cannot communicate with the brain–computer interface of real fish in real time due to the limitation of underwater communication. Therefore, a reasonable application strategy is for the operator to issue macro tasks and then transmit them to the autonomous decision-making system of the real fish. The autonomous decision-making system is the core component of the intelligent monitoring platform, and its main task is to automatically decompose and plan tasks according to the high-level instructions given by the operator, so as to realize the efficient execution of complex tasks. These systems combine task decomposition, resource management, and path planning techniques to meet operational requirements in dynamic and uncertain environments.

The main functions of the autonomous decision system can be summarized as follows:

Task decomposition: High-level task instructions are received and decomposed into operable small tasks based on the logic and operational requirements of the task. The methods used include techniques such as hierarchical task analysis (HTA) and semantic task tree to ensure the logic and feasibility of task decomposition [66].

Planning and allocation: After task decomposition is complete, the system needs to allocate resources for each small task and design an execution plan. This process may involve the introduction of optimization models such as Markov decision processes (MDPs) and Mixed-Observation MDPs (MOMDPs) for the optimal assignment of tasks and conflict avoidance in dynamic environments [67].

Path planning and execution monitoring: The execution of small tasks needs to be supported by path planning and real-time monitoring technology to ensure that tasks can be successfully completed in the face of environmental changes. For example,

path planning techniques can employ sampling algorithms and probability-based filtering methods to improve planning efficiency and reduce computational complexity [68].

In terms of technical implementation, the autonomous decision system supports efficient task planning and execution through a variety of advanced models. Behavior trees (BTs) can map abstract instructions to concrete operations to achieve dynamically adjustable task planning. In multi-robot collaboration, the distributed architecture leverages task dependencies and resource requirements to support fast task decomposition and execution, such as in, for example, fire rescues; in industrial maintenance tasks, the planning system combining CAD data and RGBD vision technology shows excellent adaptability in complex and dynamic environments [69, 70].

Similar research has been conducted in the field of animal robots. For example, Zhang et al. introduced a bird's-eye camera to monitor the experimental scene in real time and used image processing technology to track the movement information of rats (including body position and head direction) to study the decision algorithm based on a state machine for inducing robots to complete navigation tasks [71]. Secondly, Sun et al. proposed a new learning and imitation method based on a generalized regression neural network (GRNN), which replaces the traditional method of explicitly describing control logic. This method regards the navigation task as a whole. By synchronizing the control instructions of human operators with the movement information of rats, the GRNN model can automatically generate navigation instructions after a short period of training, so as to realize the autonomous navigation of robots [72]. Zhang et al. artificially enhanced the robot's exploration ability in an unknown environment by introducing a hierarchical reward generation algorithm combined with a Q-learning method of reinforcement learning to generate

progressively increasing reward signals according to the distance between the rat robot and the target. Through trial-and-error learning, the robot was able to converge on the optimal path. This demonstrates the potential of hybrid intelligent systems in complex environments [73]. These studies show that combining real-time monitoring, machine learning, and reinforcement learning can significantly improve the efficiency and adaptability of biological robots in autonomous navigation tasks, while providing new ideas for the development of hybrid intelligent systems.

Through task decomposition, resource planning, and path optimization, the autonomous decision system realizes the effective transformation and execution from large tasks to small tasks. These systems provide key technical support for improving the efficiency and adaptability of intelligent monitoring platforms and will continue to drive advances in automation technology in the future.

The fish control subsystem and the autonomous decision subsystem can constitute the central system of the whole system, and the central system consists of three parts:

Brain–computer processor: The first part is the brain–computer processor, which loads the trained deep reinforcement learning model, which needs to be introduced in the next section.

Motion-induced part: The second part is the motion-induced part; that is, the motion-related areas of the brains of the fish are connected by tiny electrodes and induced by electric discharge.

Feedback condition part: The third part is the feedback condition part, which mainly uses camera data and IMU (Inertial Measurement Unit) data; the deep reinforcement learning model will generate different electrical signals according to different environments. The specific architecture is shown in **Figure 6**.

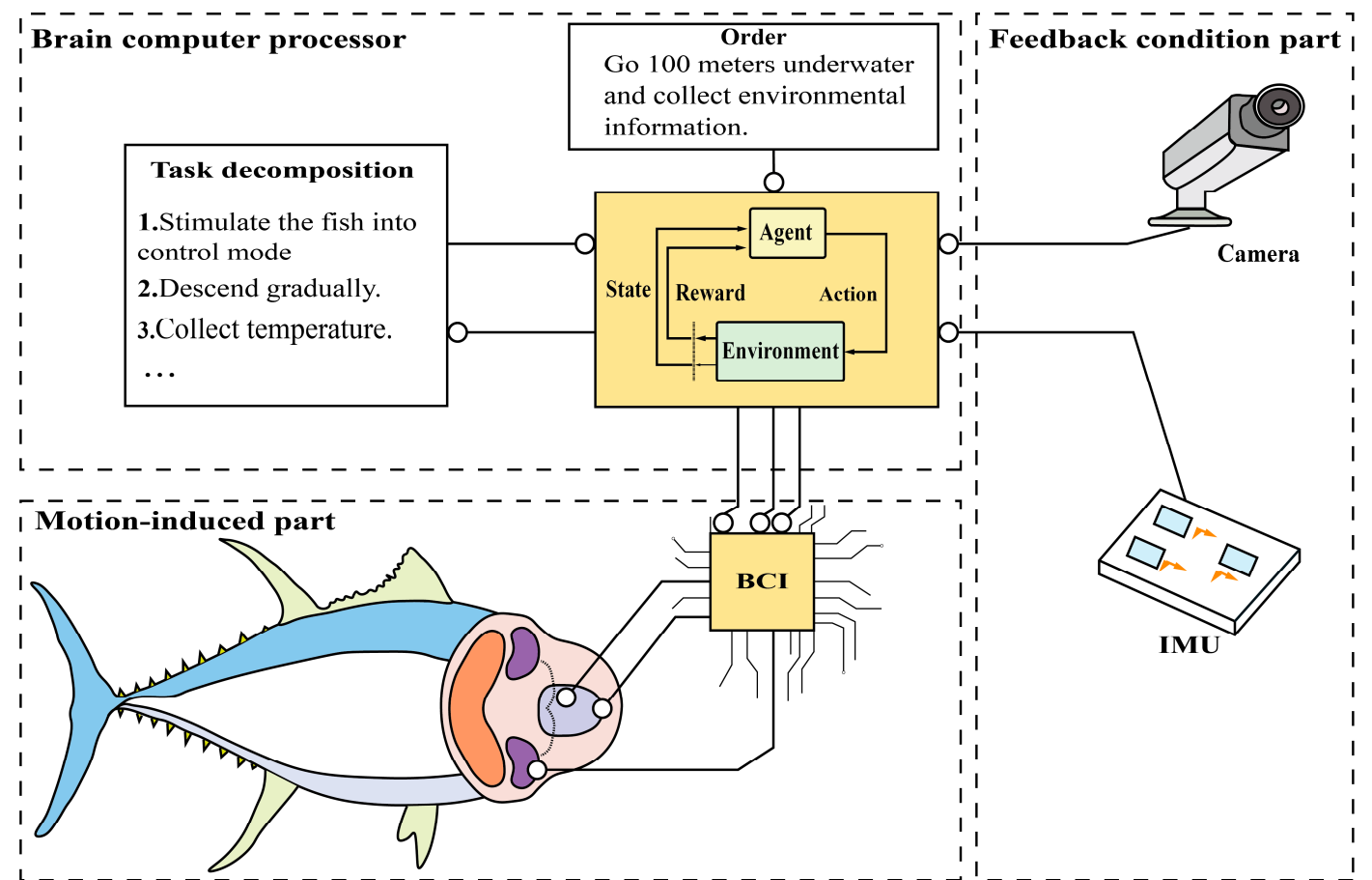


Figure 6 • Main architecture diagram of the central system.

5.3. Sensor system

The miniaturization of sensors is the main challenge for designing the sensing system. The pollution of the marine environment has become a global concern, and a marine monitoring system is urgently needed. In order to achieve a reliable and stable ocean monitoring platform, small, low-cost, high-precision sensors are essential. Optofluidic technology combines optics and microfluidics at the microscale to control light and fluids synergistically, enabling applications like integrated biosensing, adaptive optical systems, and lab-on-a-chip devices through miniaturized, reconfigurable platforms. It is speculated that the marine monitoring platform based on optofluidic sensor technology can provide accurate and

real-time monitoring of various parameters for the marine carbon cycle.

The source of marine environmental pollution is mainly the diffuse discharge of chemical substances, which is closely related to their sources. The key monitoring parameters summarized here are pH value, dissolved oxygen (DO), and heavy-metal elements [74]. Traditional sensing systems tend to be conducive to large-scale sensor arrays to achieve real-time and large-scale monitoring, but the sensor chip based on the optofluidic technology proposed in this paper can achieve compact and low-cost, real-time monitoring [75]. Optofluidic technology is an emerging field that blends optical and microfluidic technologies and has unique advantages in

improving sensing performance and system miniaturization [76]. This technique enables a highly sensitive and accurate monitoring of various optical parameters. Moreover, the materials for making optofluidic chips are relatively inexpensive, such as polydimethylsiloxane (PDMS) and polymethyl methacrylate (PMMA) [77–79]. Therefore, the advantages of optofluidic chips can be summarized as having high integration, accuracy, sensitivity, and affordability. It is believed that optofluidic chips can contribute to the field of marine monitoring in the future.

6. Conclusions

We propose an innovative concept for a fourth-generation underwater vehicle based on live fish, integrated with brain–computer interface technology, to transform live fish into a low-cost and efficient marine environment monitoring platform. By analyzing the performance and cost of existing underwater vehicles, we argue that the use of brain–computer interfaces to achieve the precise control of live fish behavior not only significantly reduces construction and operation costs, but also has great advantages in the flexibility and ecological compatibility of mission execution. At the same time, through SWOT analysis and key technology identification, this study revealed the technical feasibility and future potential of the live fish monitoring system in task decomposition, autonomous decision-making, and sensor integration. Although the current technology still faces challenges in sensor miniaturization and control system optimization, this study provided important theoretical and practical guidance for the development of future ocean monitoring technology. When using live fish as biosensors in environmental monitoring systems, we must also consider the ethical significance and practical feasibility. First, from an animal welfare perspective, strict practices need to be established to ensure that pain and stress reactions in fish are minimized during monitoring. This includes (1) optimizing sensor implantation technology to reduce tissue damage; (2) controlling and monitoring environmental parameters to maintain suitable living conditions; and (3) establishing reasonable monitoring times and work intensities. Secondly, from the perspective of practical value, a scientific evaluation system should be established to weigh the benefits of information acquisition against the costs of animal welfare. Specifically, we can (1) compare the cost-effectiveness of biological monitoring with traditional monitoring methods; (2) evaluate the data accuracy and reliability of biosensors; and (3) analyze the effects of long-term use on fish health. The application of this biological monitoring technology can be ethically and scientifically justified only if it is shown to be better than traditional methods while ensuring animal welfare. In addition, it is recommended to establish an independent ethical review committee to conduct ongoing monitoring and evaluation of relevant research.

Acknowledgments

The authors would like to thank the School of Engineering at Westlake University and the Dingheng Group for their valuable support to this work.

Funding

This research was supported by the Scientific Research Funding Project of Westlake University under Grant No. WU2024A001. The

funding was donated by Mr. Li Duo Zhu, the president of Dingheng Shipping Technology Co., Ltd.

Author contributions

Conceptualization, X.S. and W.C.; methodology, J.Y. and X.S.; software, X.S.; validation, J.Y., M.S. and W.C.; formal analysis, W.C.; investigation, X.S.; resources, X.S. and W.C.; data curation, X.S.; writing—original draft preparation, X.S. and W.C.; writing—review and editing, X.S., J.Y., M.S. and W.C.; visualization, X.S.; supervision, J.Y., M.S. and W.C.; project administration, W.C.; funding acquisition, W.C. All authors have read and agreed to the published version of the manuscript.

Conflict of interest

The authors declare no conflict of interest.

Data availability statement

Data supporting these findings are available within the article, at <https://doi.org/10.20935/AcadEng7721>, or upon request.

Institutional review board statement

Not applicable.

Informed consent statement

Not applicable.

Additional information

Received: 2024-12-12

Accepted: 2025-05-09

Published: 2025-05-21

Academia Engineering papers should be cited as *Academia Engineering* 2025, ISSN 2994-7065, <https://doi.org/10.20935/AcadEng7721>. The journal's official abbreviation is *Acad. Eng.*

Publisher's note

Academia.edu Journals stays neutral with regard to jurisdictional claims in published maps and institutional affiliations. All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

Copyright

© 2025 copyright by the authors. This article is an open access article distributed under the terms and conditions of the Creative

Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

References

1. Council NR, Earth Do, Studies L, Board OS. Priorities CoGfNoNOSR, sciences DSoO. In: Sea change: 2015–2025 decadal survey of ocean sciences. Washington (DC): National Academies Press; 2015.
2. Li J, Li W, Liu Q, Luo B, Cui W. Current status and technical challenges in the development of biomimetic robotic fish-type submersible. *Ocean-Land-Atmos Res*. 2024;3:0036. doi: 10.34133/olar.0036
3. Triantafyllou MS, Triantafyllou GS. An efficient swimming machine. *Sci Am*. 1995;272(3):64–70. doi: 10.1038/scientificamerican0395-64
4. Ma S, Zhao Q, Ding M, Zhang M, Zhao L, Huang C, et al. A review of robotic fish based on smart materials. *Biomimetics*. 2023;8(2):227. doi: 10.3390/biomimetics8020227
5. Sun B, Li W, Wang Z, Zhu Y, He Q, Guan X, et al. Recent progress in modeling and control of bio-inspired fish robots. *J Mar Sci Eng*. 2022;10(6):773. doi: 10.3390/jmse10060773
6. Salhaoui M, Molina-Molina JC, Guerrero-González A, Arioua M, Ortiz FJ. Autonomous underwater monitoring system for detecting life on the seabed by means of computer vision cloud services. *Remote Sens*. 2020;12(12):1981. doi: 10.3390/rs12121981
7. Liu Z, Zhang S, Yang C, Chung W-H, Li Z. Submarine optical fiber sensing system for the real-time monitoring of depth, vibration, and temperature. *Front Mar Sci*. 2022;9:922669. doi: 10.3389/fmars.2022.922669
8. Long R. Problems and significance of marine environment monitoring. *E3S Web of Conf*. 2023;393:01005. doi: 10.1051/e3sconf/202339301005
9. Yang S, Liu P, Lim TH. IoT-based underwater robotics for water quality monitoring in aquaculture: A survey. *Proceedings of the 11th International Conference on Robot Intelligence Technology and Applications (RiTA 2023)*; 2023 December 6–8; Taicang, China. Berlin: Springer; 2023.
10. Bayat B, Crasta N, Crespi A, Pascoal AM, Ijspeert A. Environmental monitoring using autonomous vehicles: a survey of recent searching techniques. *Curr Opin Biotechnol*. 2017;45:76–84. doi: 10.1016/j.copbio.2017.01.009
11. Cui W, Pan L, Li R. A suggestion of using task efficiency to replace swimming efficiency for both robotic fish and living fish. *Ships Offshore Struct*. 2024;19:2204–12. doi: 10.1080/17445302.2024.2331301
12. Bale R, Hao M, Bhalla APS, Patankar NA. Energy efficiency and allometry of movement of swimming and flying animals. *Proc Natl Acad Sci USA*. 2014;111(21):7517–21. doi: 10.1073/pnas.1310544111
13. Bale R, Hao M, Bhalla APS, Patel N, Patankar NA. Gray's paradox: a fluid mechanical perspective. *Sci Rep*. 2014;4(1):5904. doi: 10.1038/srep05904
14. Gray J. Studies in animal locomotion: VI. The propulsive powers of the dolphin. *J Exp Biol*. 1936;13(2):192–9. doi: 10.1242/jeb.13.2.192
15. Maertens AP, Triantafyllou MS, Yue DK. Efficiency of fish propulsion. *Bioinspiration Biomim*. 2015;10(4):046013. doi: 10.1088/1748-3190/10/4/046013
16. Schultz WW, Webb PW. Power requirements of swimming: do new methods resolve old questions? *Integr Comp Biol*. 2002;42(5):1018–25. doi: 10.1093/icb/42.5.1018
17. Bruslé J, Quignard J-P. *Fish Behavior 1: eco-ethology*. Hoboken (NJ): John Wiley & Sons; 2020.
18. Magnhagen C, Magurran AE. Decision-making and trade-offs in fish behaviour. In: *Fish Behaviour*. Boca Raton (FL): CRC Press; 2008. p. 499–523.
19. Kooyman GL. Techniques used in measuring diving capacities of Weddell seals. *Polar Record*. 1965;12(79):391–4. doi: 10.1017/S003224740005484x
20. Yoda K, Sato K, Niizuma Y, Kurita M, Bost C-A, Maho YL, et al. Precise monitoring of porpoising behaviour of Adélie penguins determined using acceleration data loggers. *J Exp Biol*. 1999;202(22):3121–6. doi: 10.1242/jeb.202.22.3121
21. Klimley AP, Le Boeuf BJ, Cantara KM, Richert JE, Davis SF, Van Sommeran S, et al. The hunting strategy of white sharks (*Carcharodon carcharias*) near a seal colony. *Mar Biol*. 2001;138:617–36. doi: 10.1007/s002270000489
22. McMahon CR, Autret E, Houghton JD, Lovell P, Myers AE, Hays GC. Animal-borne sensors successfully capture the real-time thermal properties of ocean basins. *Limnol Oceanogr Methods*. 2005;3(9):392–8. doi: 10.4319/lom.2005.3.392
23. Watanabe YY, Papastamatiou YP. Biologging and biotelemetry: tools for understanding the lives and environments of marine animals. *Annu Rev Anim Biosci*. 2023;11(1):247–67. doi: 10.1146/annurev-animal-050322-073657
24. Peng Y, Zhang H, Zhao Y, Zhao Z, Wen Y, Han L. Research status and progress of aquatic animal robots. *Chin J Biomed Eng*. 2023;42(5):610–6. doi: 10.3969/j.issn.0258-8021.2023.05.011
25. Brown S. Stealth sharks to patrol the high seas. *New Sci*. 2006;189(2541):30–321.
26. Lehmkuhle M, Vetter R, Parikh H, Carrier J, Kipke D. Implantable neural interfaces for characterizing population responses to odorants and electrical stimuli in the nurse shark, *Ginglymostoma cirratum*. In: *Chemical Senses*. Oxford: Oxford Univ Press; 2006.
27. Kim C-H, Choi B, Kim D-G, Lee S, Jo S, Lee P-S. Remote navigation of turtle by controlling instinct behavior via human brain-computer interface. *J Bionic Eng*. 2016;13(3):491–503. doi: 10.1016/s1672-6529(16)60322-0
28. Kim D-G, Lee S, Kim C-H, Jo S, Lee P-S. Parasitic robot system for waypoint navigation of turtle. *J Bionic Eng*. 2017;14(2):327–35. doi: 10.1016/s1672-6529(16)60401-8

29. Peng Y, Wang T, Yan Y, Chen Z, Wen S, Han X, et al. Design and application of wireless remote control system for carp robot. *Chinese J Bionic Eng.* 2019;38(4):431–7. doi: 10.3969/j.issn.0258-8021.2019.04.7
30. Peng Y, Zhao Y, Zhang Q, Bao Z, Liu X, Zhang H, et al. Study on the influence of different wavelengths of light on the motion behavior of carp robot. *J Bionic Eng.* 2021;38(4):647. doi: 10.7507/1001-5515.202012078
31. Kobayashi N, Yoshida M, Matsumoto N, Uematsu K. Artificial control of swimming in goldfish by brain stimulation: confirmation of the midbrain nuclei as the swimming center. *Neurosci Lett.* 2009;452(1):42–6. doi: 10.1016/j.neulet.2009.01.035
32. Vinepinsky E, Cohen L, Perchik S, Ben-Shahar O, Donchin O, Segev R. Representation of edges, head direction, and swimming kinematics in the brain of freely-navigating fish. *Sci Rep.* 2020;10(1):14762. doi: 10.1101/291013
33. Vinepinsky E, Donchin O, Segev R. Wireless electrophysiology of the brain of freely swimming goldfish. *J Neurosci Methods.* 2017;278:76–86. doi: 10.1016/j.jneumeth.2017.01.001
34. Talwar SK, Xu S, Hawley ES, Weiss SA, Moxon KA, Chapin JK. Rat navigation guided by remote control. *Nature.* 2002;417(6884):37–8. doi: 10.1038/417037a
35. Huai R, Yang J, Wang H, Su X. A new robo-animals navigation method guided by the remote control. *Proceedings of the 2009 2nd International Conference on Biomedical Engineering and Informatics*; 2009 October 17–19; Tianjin, China. Piscataway (NJ): IEEE; 2009.
36. Xu K, Zhang J, Zhou H, Lee JCT, Zheng X. A novel turning behavior control method for rat-robot through the stimulation of ventral posteromedial thalamic nucleus. *Behav Brain Res.* 2016;298:150–7. doi: 10.1016/j.bbr.2015.11.002
37. Chen S, Qu Y, Guo S, Shi Z, Xu K, Zheng X. Encode the “STOP” command by photo-stimulation for precise control of rat-robot. *Proceedings of the 2013 35th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*; 2013 July 3–7; Osaka, Japan. Piscataway (NJ): IEEE; 2013.
38. Lin J-Y, Yu C-N, Jia J, Zhang S-M, Wang Y-W, Chen W-D, et al. Using dIPAG-evoked immobile behavior in animal-robotics navigation. *Proceedings of the 2010 5th International Conference on Computer Science & Education*; 2010 August 24–27; Hefei, China. Piscataway (NJ): IEEE; 2010.
39. Caggiano V, Leiras R, Goni-Erro H, Masini D, Bellardita C, Bouvier J, et al. Midbrain circuits that set locomotor speed and gait selection. *Nature.* 2018;553(7689):455–60. doi: 10.1038/nature25448
40. Capelli P, Pivetta C, Soledad Esposito M, Arber S. Locomotor speed control circuits in the caudal brainstem. *Nature.* 2017;551(7680):373–7. doi: 10.1038/nature24064
41. Su X, Huai R, Yang J, Wang H, Lv C. Brain mechanisms and control methods for controlling motor behavior of animal robots. *Sci China Inf Sci.* 2012;42(9):1130–46. doi: 10.1360/112012-522
42. Cai L, Dai Z, Wang W, Wang H, Tang Y. Modulating motor behaviors by electrical stimulation of specific nuclei in pigeons. *J Bionic Eng.* 2015;12(4):555–64. doi: 10.1016/s1672-6529(14)60145-1
43. Dong M, Cai L, Wang H, Dai Z, Wang W. A preliminary study on motion induced by electric stimulation of the midbrain in pigeons. *J Biolog.* 2012;29(4):33–5.
44. Bozkurt A, Lal A, Gilmour R. Aerial and terrestrial locomotion control of lift assisted insect biobots. *Proceedings of the 2009 Annual International Conference of the IEEE Engineering in Medicine and Biology Society*; 2009 September 3–6; Minneapolis, MN, USA. Piscataway (NJ): IEEE; 2009.
45. Latif T, Bozkurt A. Line following terrestrial insect biobots. *Proceedings of the 2012 Annual International Conference of the IEEE Engineering in Medicine and Biology Society*; 2012 August 28–September 1; San Diego, CA, USA. Piscataway (NJ): IEEE; 2012.
46. Whitmire E, Latif T, Bozkurt A. Kinect-based system for automated control of terrestrial insect biobots. *Proceedings of the 2013 35th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*; 2013 July 3–7; Osaka, Japan. Piscataway (NJ): IEEE; 2013.
47. Sato H, Peeri Y, Baghoomian E, Berry C, Maharbiz M. Radio-controlled cyborg beetles: a radio-frequency system for insect neural flight control. *Proceedings of the 2009 IEEE 22nd International Conference on Micro Electro Mechanical Systems*; 2009 January 25–29; Sorrento, Italy. Piscataway (NJ): IEEE; 2009.
48. Zheng N, Jin M, Hong H, Huang L, Gu Z, Li H. Real-time and precise insect flight control system based on virtual reality. *Electron Lett.* 2017;53(6):387–9. doi: 10.1049/el.2016.3048
49. Benzaghta MA, Elwalda A, Mousa MM, Erkan I, Rahman M. SWOT analysis applications: an integrative literature review. *J Glob Bus Insights.* 2021;6(1):54–72. doi: 10.5038/2640-6489.6.1.1148
50. Tavakoli S, Khojasteh D, Haghani M, Hirdaris S. A review on the progress and research directions of ocean engineering. *Ocean Eng.* 2023;272:113617. doi: 10.1016/j.oceaneng.2023.113617
51. Treasure AM, Roquet F, Ansoorge IJ, Bester MN, Boehme L, Bornemann H, et al. Marine mammals exploring the oceans pole to pole: a review of the MEOP consortium. *Oceanography.* 2017;30(2):132–8. doi: 10.5670/oceanog.2017.234
52. Zhao Y, Peng Y, Wen Y, Han L, Zhang H, Zhao Z, et al. A novel movement behavior control method for carp robot through the stimulation of medial longitudinal fasciculus nucleus of midbrain. *J Bionic Eng.* 2022;19(5):1302–13. doi: 10.1007/s42235-022-00211-2
53. Kotrschal A, Kotrschal K. Fish brains: anatomy, functionality, and evolutionary relationships. *Welf Fish.* 2020:129–48. doi: 10.1007/978-3-030-41675-1_6
54. Saitoh K, Ménard A, Grillner S. Tectal control of locomotion, steering, and eye movements in lamprey. *J Neurophysiol.* 2007;97(4):3093–108. doi: 10.1152/jn.00639.2006

55. Zelenin PV. Reticulospinal neurons controlling forward and backward swimming in the lamprey. *J Neurophysiol.* 2011;105(3):1361–71. doi: 10.1152/jn.00887.2010
56. Green MH, Hale ME. Activity of pectoral fin motoneurons during two swimming gaits in the larval zebrafish (*Danio rerio*) and localization of upstream circuit elements. *J Neurophysiol.* 2012;108(12):3393–402. doi: 10.1152/jn.00623.2012
57. Roberts B, Russell I. The activity of lateral-line efferent neurones in stationary and swimming dogfish. *J Exp Biol.* 1972; 57(2):435–48. doi: 10.1242/jeb.57.2.435
58. Musk E. An integrated brain-machine interface platform with thousands of channels. *J Med Internet Res.* 2019;21(10): e16194. doi: 10.2196/16194
59. Gilja V, Nuyujukian P, Chestek CA, Cunningham JP, Byron MY, Fan JM, et al. A brain machine interface control algorithm designed from a feedback control perspective. Proceedings of the 2012 Annual International Conference of the IEEE Engineering in Medicine and Biology Society; 2012 August 28–September 1; San Diego, CA, USA. Piscataway (NJ): IEEE; 2012.
60. Li X, Duraisamy K, Bogdan P, Doppa JR, Pande PP. Scalable network-on-chip architectures for brain-machine interface applications. *IEEE Trans Very Large Scale Integr VLSI Syst.* 2018;26(10):1895–907. doi: 10.1109/tvlsi.2018.2843282
61. Mercanzini A, Cheung K, Buhl DL, Boers M, Maillard A, Colin P, et al. Demonstration of cortical recording using novel flexible polymer neural probes. *Sens Actuators A Phys.* 2008;143(1):90–6. doi: 10.1016/j.sna.2007.07.027
62. Bablani A, Edla DR, Tripathi D, Cheruku R. Survey on brain-computer interface: an emerging computational intelligence paradigm. *ACM Comput Surv CSUR.* 2019;52(1):1–32. doi: 10.1145/3297713
63. Patrick E, Sankar V, Rowe W, Sanchez JC, Nishida T. Design of an implantable intracortical microelectrode system for brain-machine interfaces. Proceedings of the 2009 4th International IEEE/EMBS Conference on Neural Engineering; 2009 April 29–May 2; Antalya, Turkey. Piscataway (NJ): IEEE; 2009.
64. Liu X, Zhang M, Subei B, Richardson AG, Lucas TH, Van der Spiegel J. The PennBMBI: Design of a general purpose wireless brain-machine-brain interface system. *IEEE Trans Biomed Circuits Syst.* 2015;9(2):248–58. doi: 10.1109/tb-cas.2015.2392555
65. Zhao Y, Peng Y, Wen Y, Han L, Yan Y, Dong X, et al. Identification of Motor Nuclei in the Medulla Oblongata of Carp for Biological Control. *J Bionic Eng.* 2024;21(2):842–51. doi: 10.1007/s42235-023-00456-5
66. Ren W, Yang X, Yan Y, Hu Y. The decision-making framework for assembly tasks planning in human-robot collaborated manufacturing system. *Int J Comput Integr Manuf.* 2023;36(2):289–307. doi: 10.1080/0951192x.2022.2081359
67. Chen Y, Rosolia U, Ames AD. Decentralized task and path planning for multi-robot systems. *IEEE Robot Autom Lett.* 2021;6(3):4337–44. doi: 10.1109/lra.2021.3068103
68. Friedrich C, Csiszar A, Lechler A, Verl A. Efficient task and path planning for maintenance automation using a robot system. *IEEE Trans Autom Sci Eng.* 2017;15(3):1205–15. doi: 10.1109/tase.2017.2759814
69. El Shenawy A, Mohamed K, Harb H. Hdec-posmdps mrs exploration and fire searching based on iot cloud robotics. *Int J Autom Comput.* 2020;17(3):364–77. doi: 10.1007/s11633-019-1187-6
70. Zhou H, Min H, Lin Y. An autonomous task algorithm based on behavior trees for robot. Proceedings of the 2019 2nd China Symposium on Cognitive Computing and Hybrid Intelligence (CCHI); 2019 September 21–22; Xi'an, China. Piscataway (NJ): IEEE; 2019.
71. Zhang Y, Sun C, Zheng N, Zhang S, Lin J, Chen W, et al. An automatic control system for ratbot navigation. Proceedings of the 2010 IEEE/ACM Int'l Conference on Green Computing and Communications & Int'l Conference on Cyber, Physical and Social Computing; 2010 December 18–20; Washington, DC, USA. Piscataway (NJ): IEEE; 2010.
72. Sun C, Zheng N, Zhang X, Chen W, Zheng X. An automatic control model for rat-robot. Proceedings of the 2011 Annual International Conference of the IEEE Engineering in Medicine and Biology Society; 2011 August 30–September 3; Boston, MA, USA. Piscataway (NJ): IEEE; 2011.
73. Zhang C, Sun C, Gao L, Zheng N, Chen W, Zheng X. Bio-robots automatic navigation with graded electric reward stimulation based on reinforcement learning. Proceedings of the 2013 35th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC); 2013 July 3–7; Osaka, Japan. Piscataway (NJ): IEEE; 2013.
74. Briciu-Burghina C, Power S, Delgado A, Regan F. Sensors for coastal and ocean monitoring. *Annu Rev Anal Chem.* 2023;16(1):451–69. doi: 10.1146/annurev-anchem-091922-085746
75. Wang F, Zhu J, Chen L, Zuo Y, Hu X, Yang Y. Autonomous and in situ ocean environmental monitoring on optofluidic platform. *Micromachines.* 2020;11(1):69. doi: 10.3390/mi11010069
76. Yung KY, Schadock-Hewitt AJ, Hunter NP, Bright FV, Baker GA. 'Liquid litmus': chemosensory pH-responsive photonic ionic liquids. *Chem Commun.* 2011;47(16):4775–7. doi: 10.1039/c1cc00065a
77. Ali U, Karim KJBA, Buang NA. A review of the properties and applications of poly (methyl methacrylate)(PMMA). *Polym Rev.* 2015;55(4):678–705. doi: 10.1080/15583724.2015.1031377
78. Wolf MP, Salieb-Beugelaar GB, Hunziker P. PDMS with designer functionalities—properties, modifications strategies, and applications. *Prog Polym Sci.* 2018;83:97–134. doi: 10.1016/j.progpolymsci.2018.06.001
79. Zafar MS. Prosthodontic applications of polymethyl methacrylate (PMMA): an update. *Polymers.* 2020;12(10): 2299. doi: 10.3390/polym12102299